Comparative Analysis of Binary Black Hole Models in the Intermediate Mass Ratio Regime

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Massachusetts Undergraduate Research Conference

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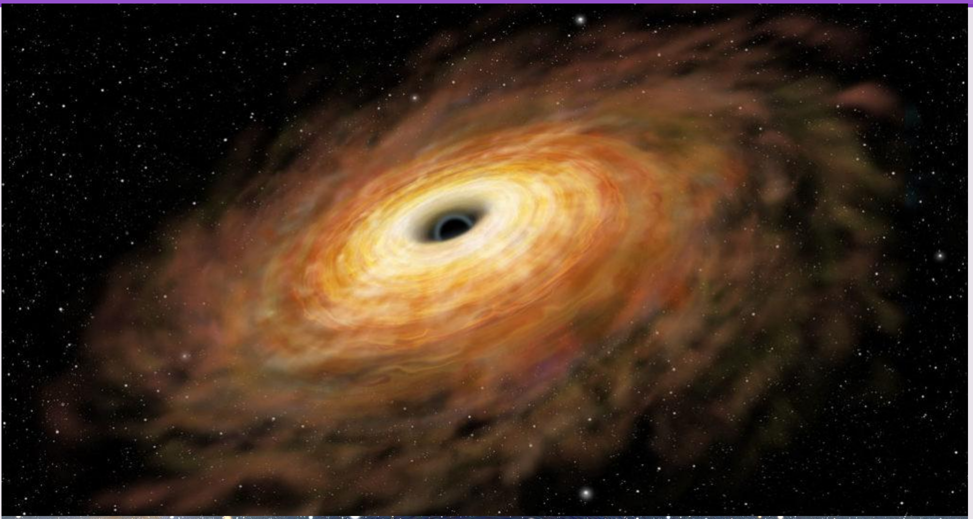
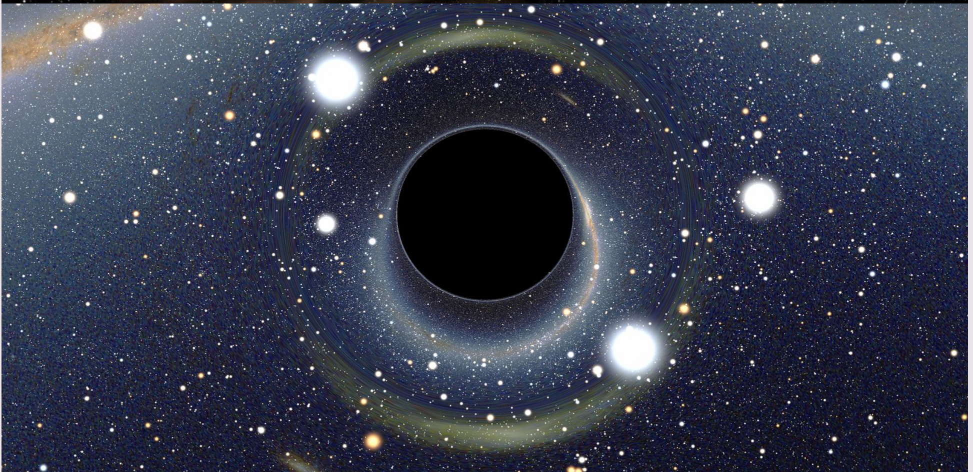
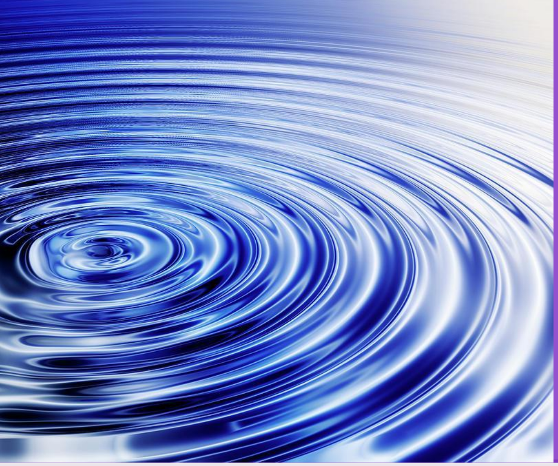
Background

Types of Black Holes



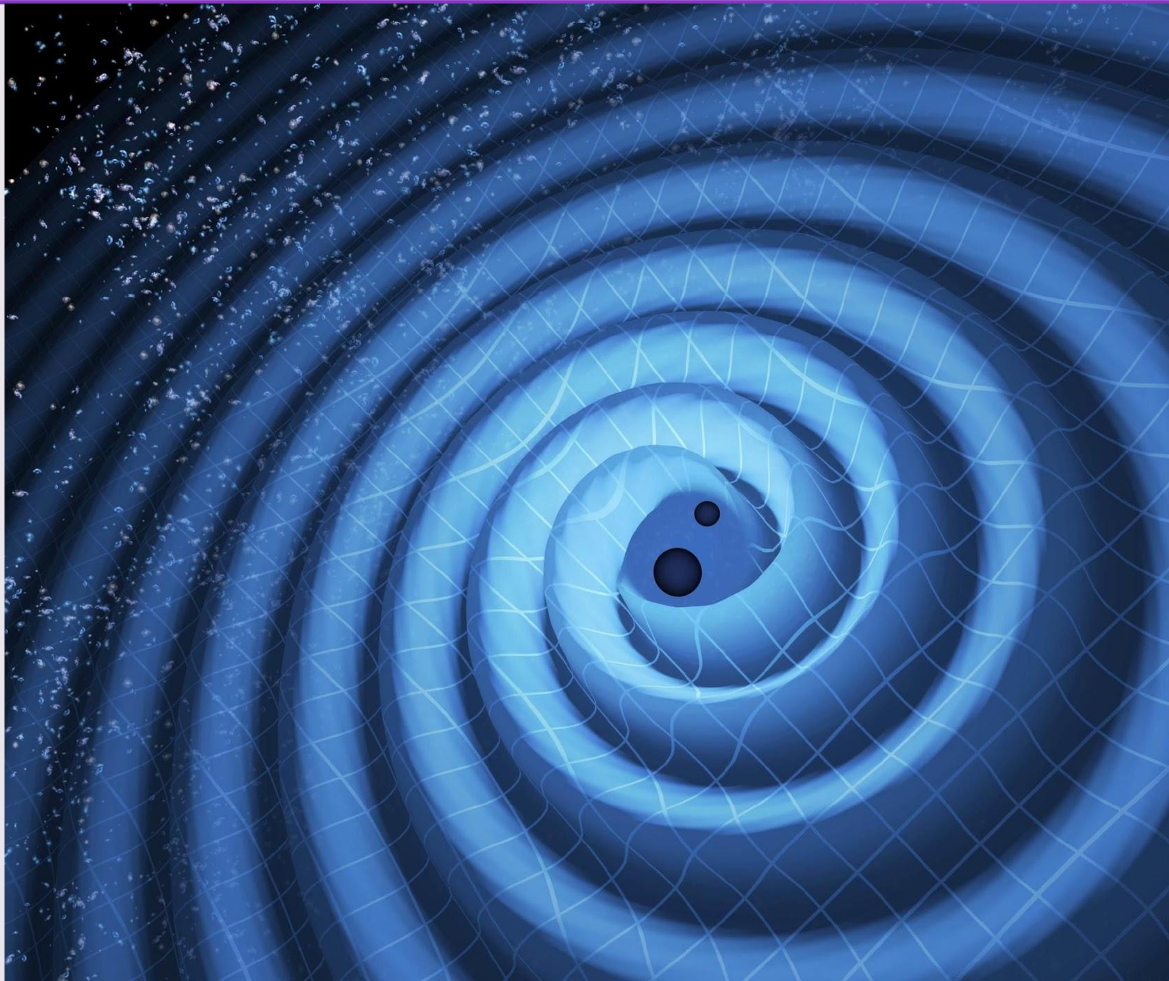


- Black holes characterized by **mass,** spin, and charge
- Stellar-mass (2.5-100 Solar Masses)
  - Star explosions
- Supermassive ($\geq$ 1,000,000 SM)
  - Center of galaxies
- **Intermediate** (100-10,000+ SM)
  - Partially unknown origins
  - Multiple stellar black hole collisions



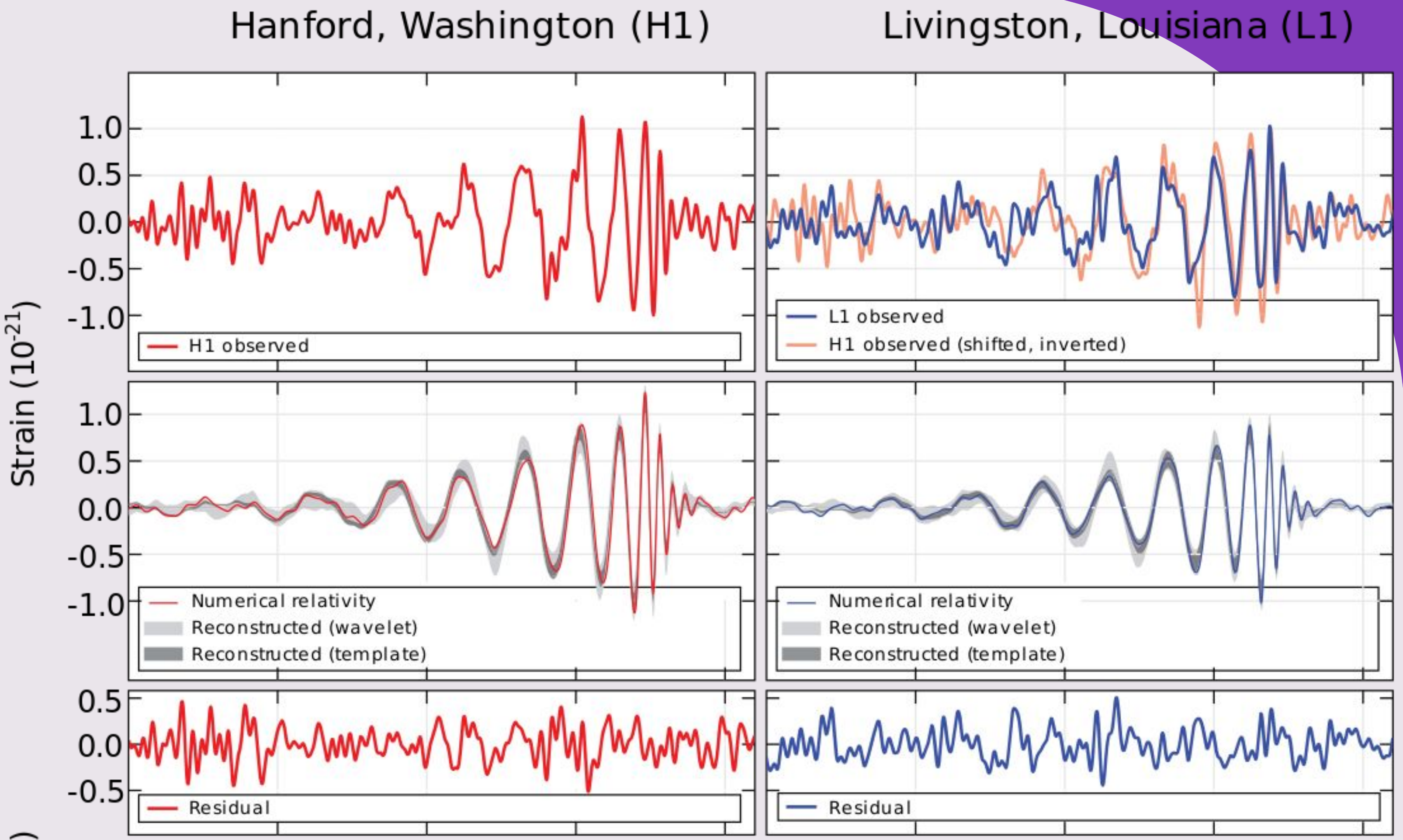
What are Gravitational Waves?

- Predicted by Einstein with General Relativity
- Caused by accelerating massive bodies
- Analogous to a ball moving along the surface of water
- Binary black hole systems (BBH)

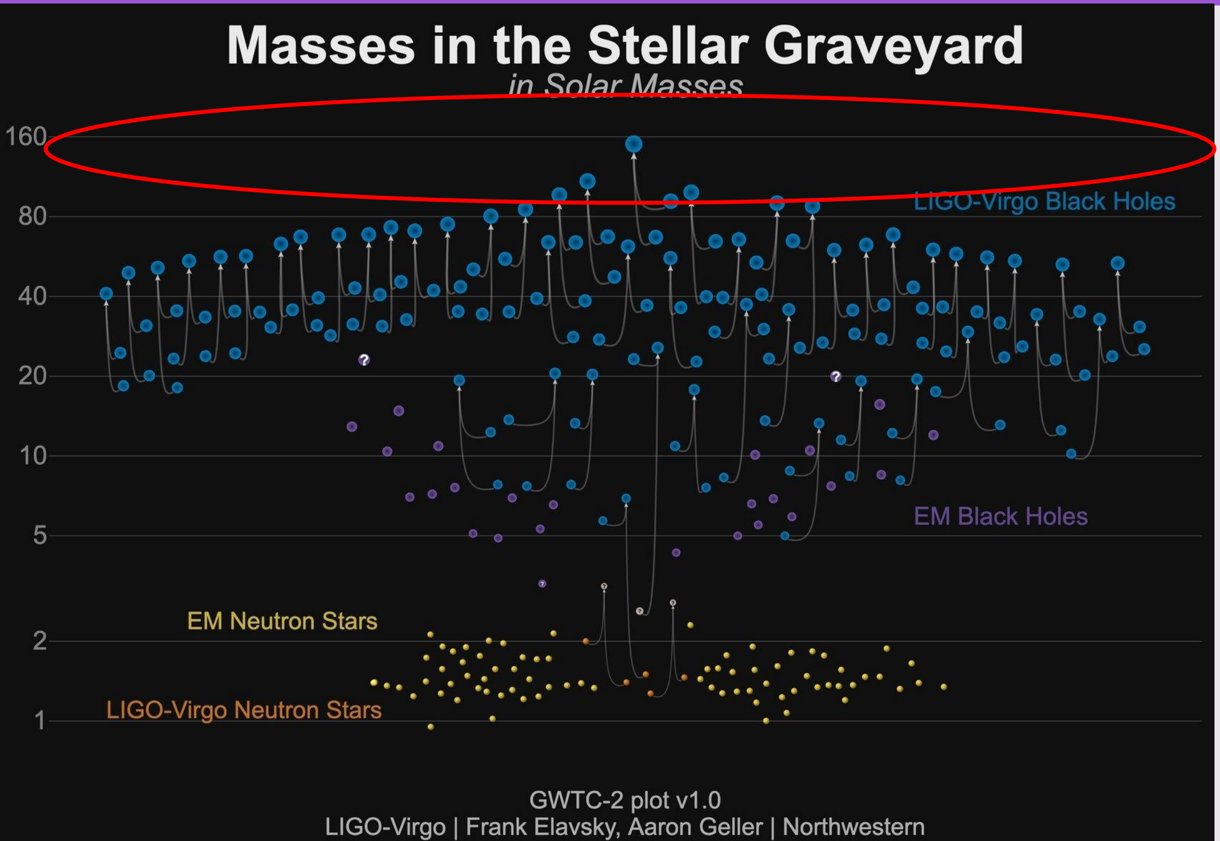


LIGO

- Laser Interferometer Gravitational-Wave Observatory
- Louisiana, Washington (VIRGO, Italy)
- Basic ideas of interferometry circa 1960's
- First detection in September 2015
- Kip Thorne, Barry Barish, and Rai Weiss won the 2018 Nobel prize for founding LIGO



New Era of Gravitational Science



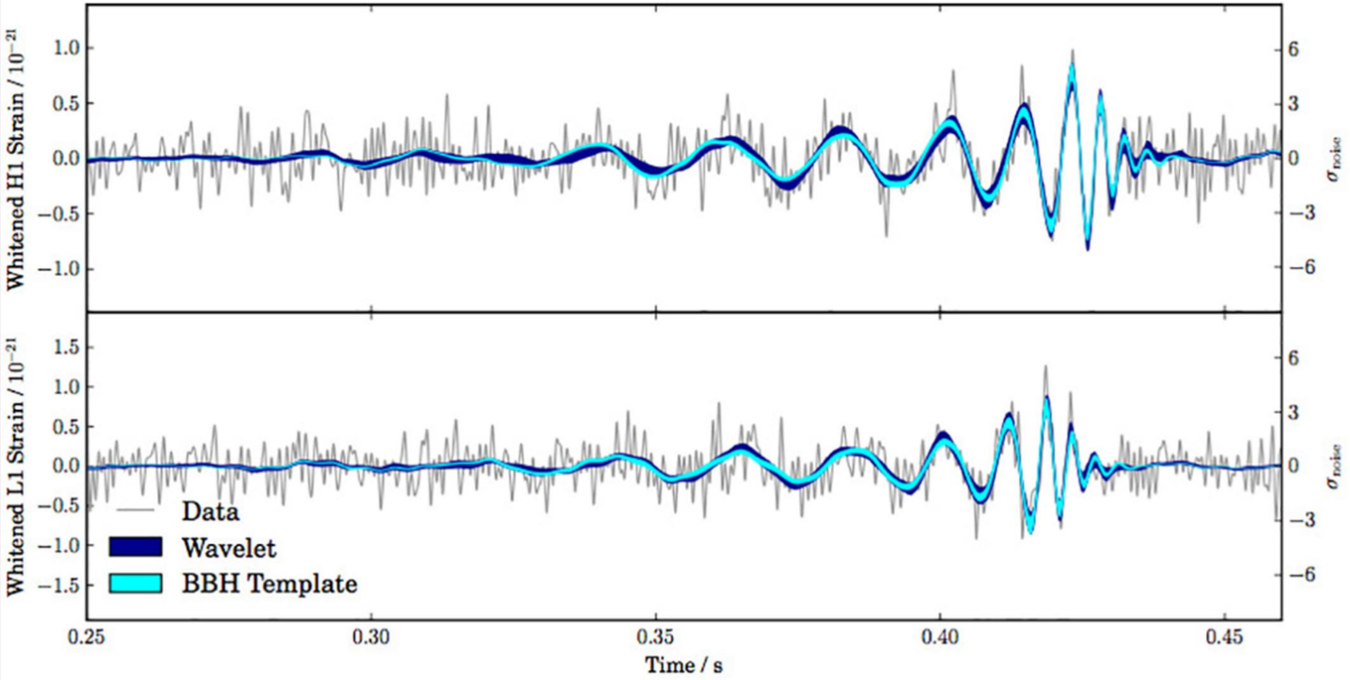
- Tests of General Relativity
- Potential probe of Quantum Gravity
- Space Interferometers in discussion
- Learning about populations of previously unseen black holes
- Very recent, entirely new tool for exploring the universe; has and will continue to bring surprising discoveries

GW190521

- Gravitational Wave detected 21 May 2019
- Merging of two BH (66, 85 SM)
- Resulting BH 142 SM
- Places new BH into Intermediate Mass regime!
- Test models against real data

Gravitational Wave Model Comparisons

Gravitational Wave Models



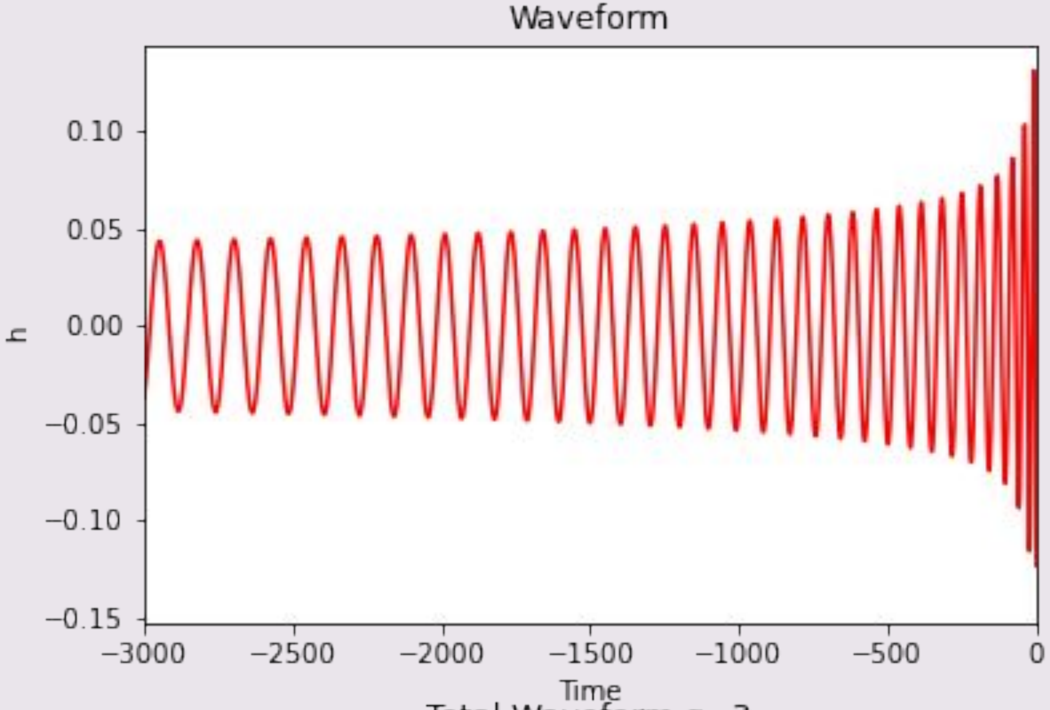
- Used to determine black hole parameters post GW detection
  - Mass, spin, location, etc.
- Model predicts waveform shape
- Real data is compared to model data
- If model disagreement > 0.01, model is inaccurate:
  - Incorrect tests of GR
  - Incorrect parameters of BBH systems

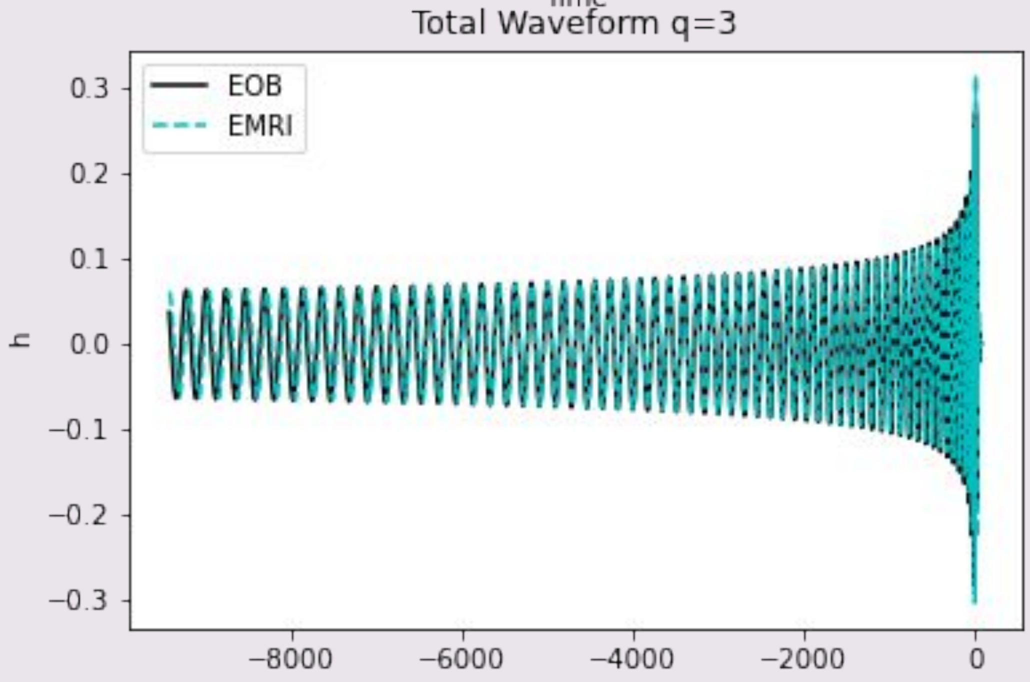
Models/ Terminology

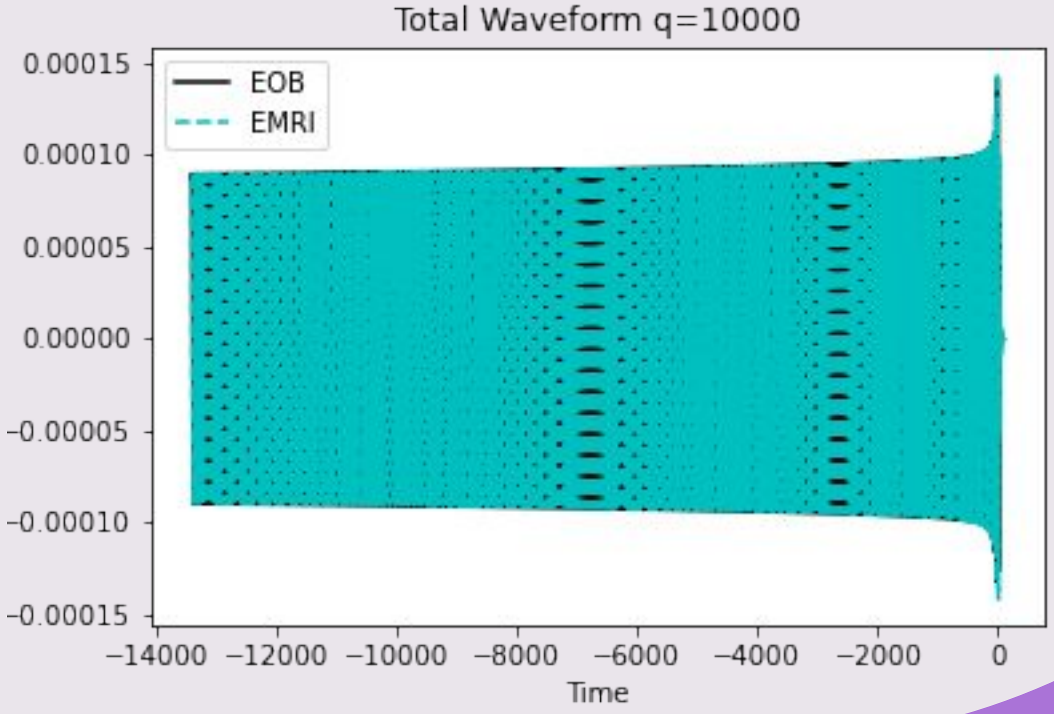
- EOB – Effective One Body Model
- LAL – LIGO Algorithm Library
  - Where EOB is obtained
- EMRI – Extreme Mass Ratio Inspiral
  - Developed by researchers at UMassD and collaborators

- q – Mass ratio (m_1/m_2)
- Mode (L, M)
  - L,M – Spherical Harmonic index values
  - Used to represent gravitational fields
  - Analogous to guitar strings

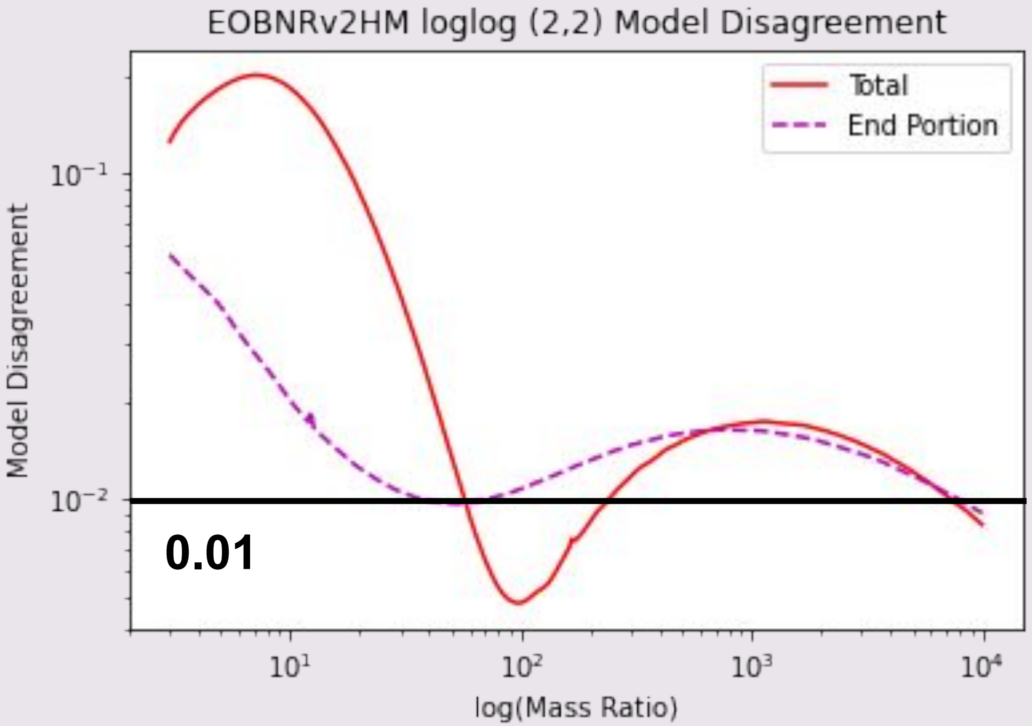
Example Waveforms

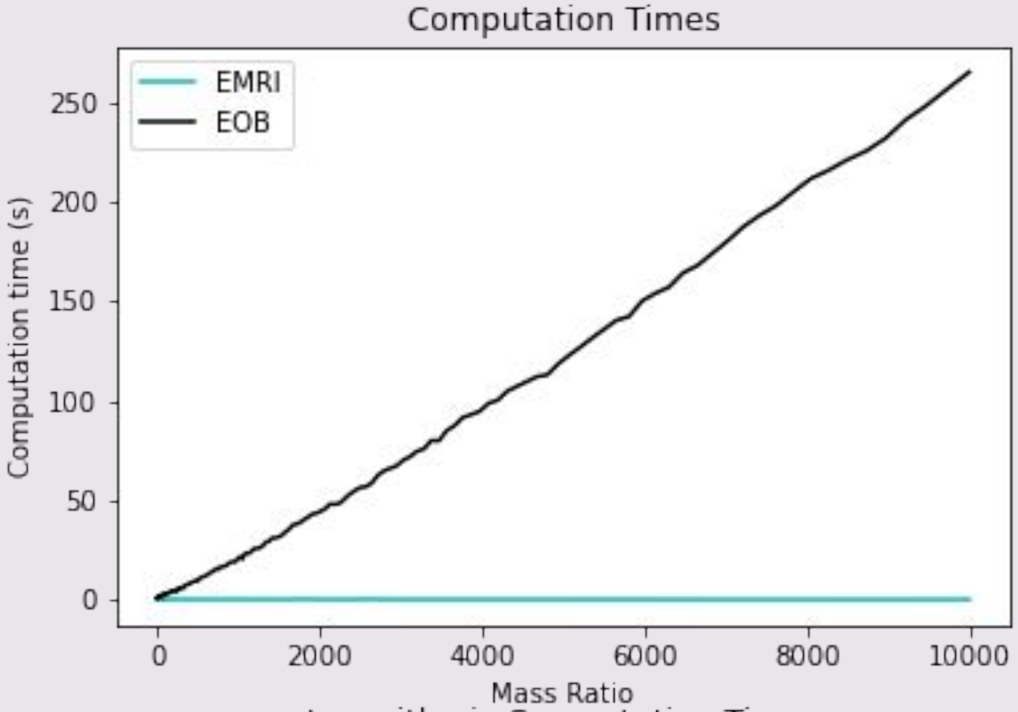


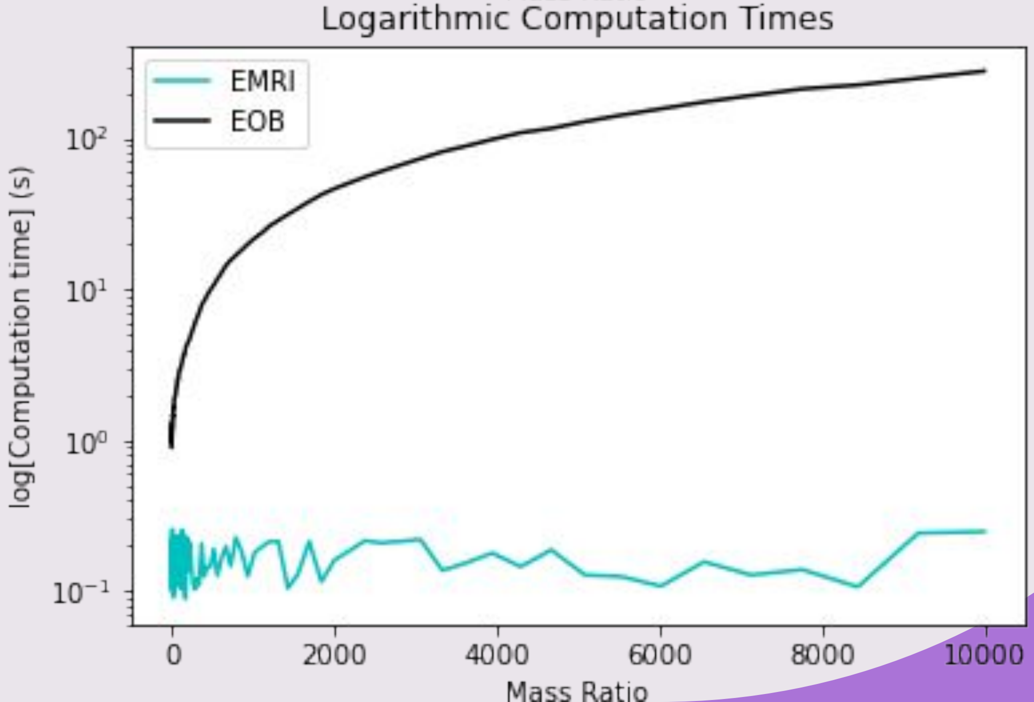




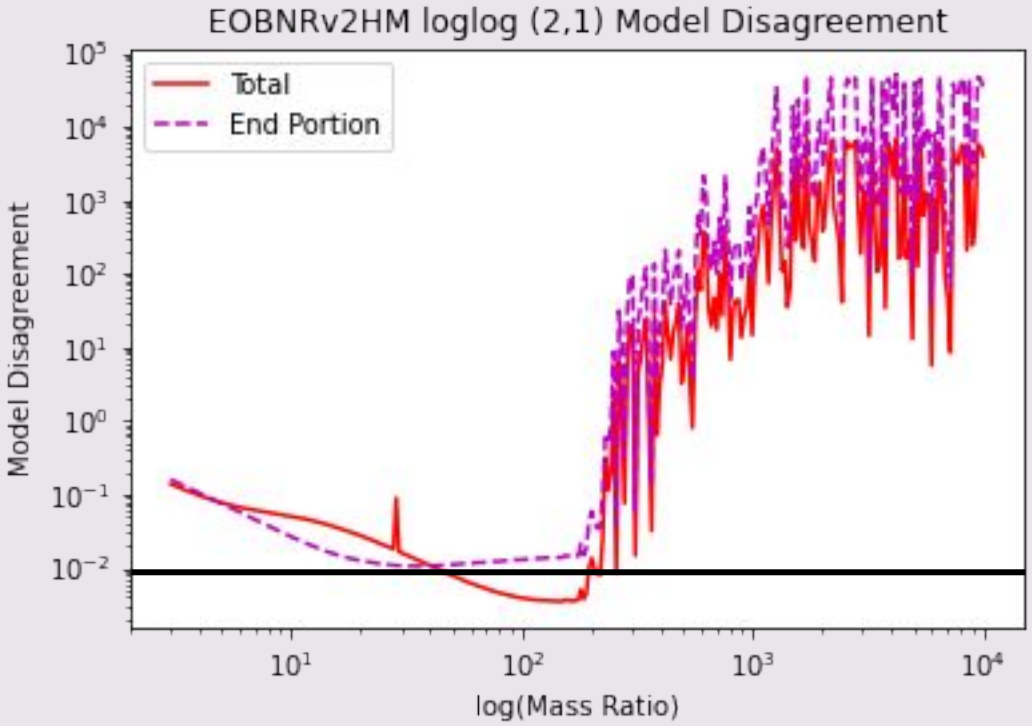
Results


EOBNRv2HM loglog (2,2) Model Disagreement
Total
End Portion
Model Disagreement
10^{-1}
10^{-2}
0.01
10^{1}
10^{2}
10^{3}
10^{4}
log(Mass Ratio)

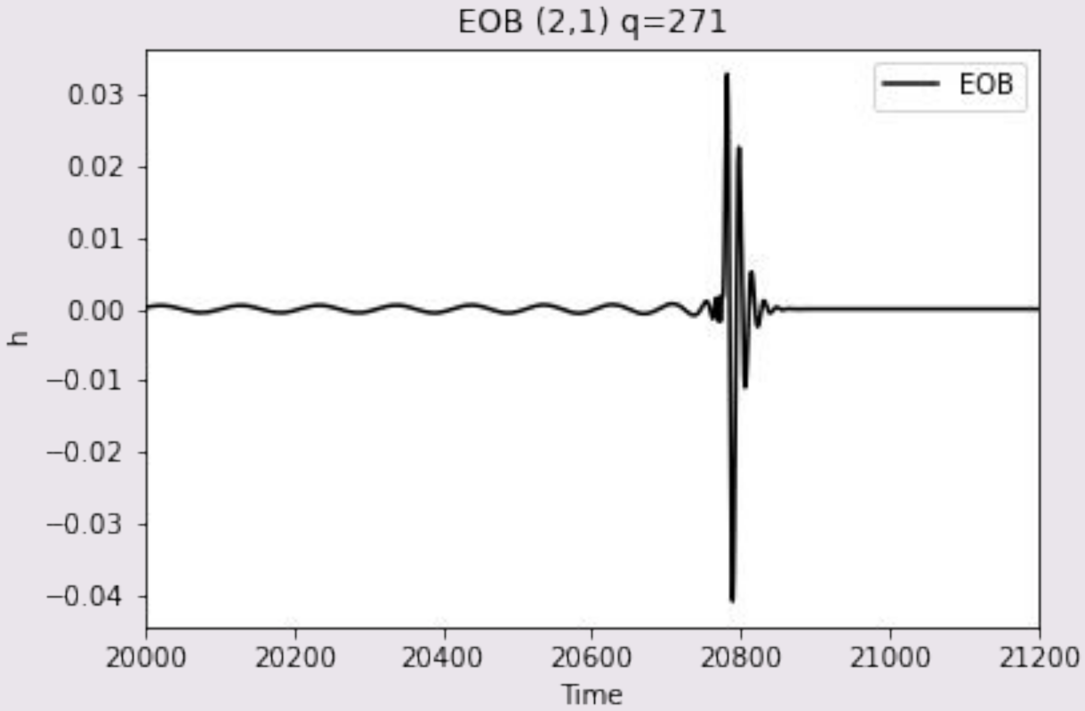

Computation Times
EMRI
EOB
Computation time (s)
250
200
150
100
50
0
0
2000
4000
6000
8000
10000
Mass Ratio


Logarithmic Computation Times
EMRI
EOB
log[Computation time] (s)
10^{2}
10^{1}
10^{0}
10^{-1}
0
2000
4000
6000
8000
10000
Mass Ratio

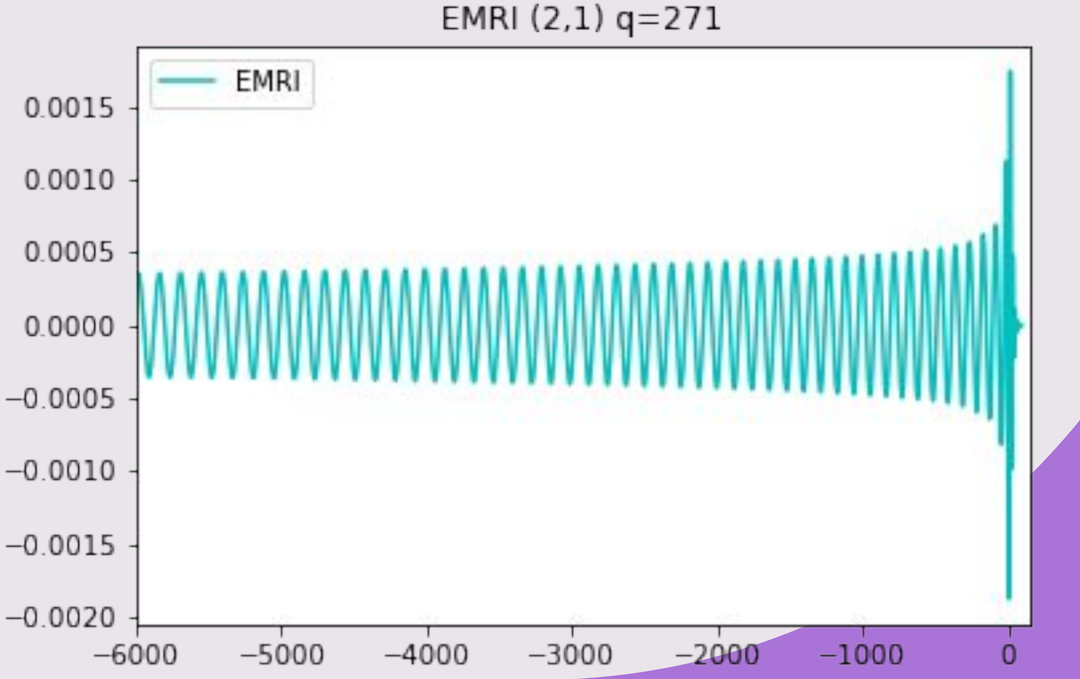
Results



EOBNRv2HM loglog (2,1) Model Disagreement
Total
End Portion
Model Disagreement
log(Mass Ratio)

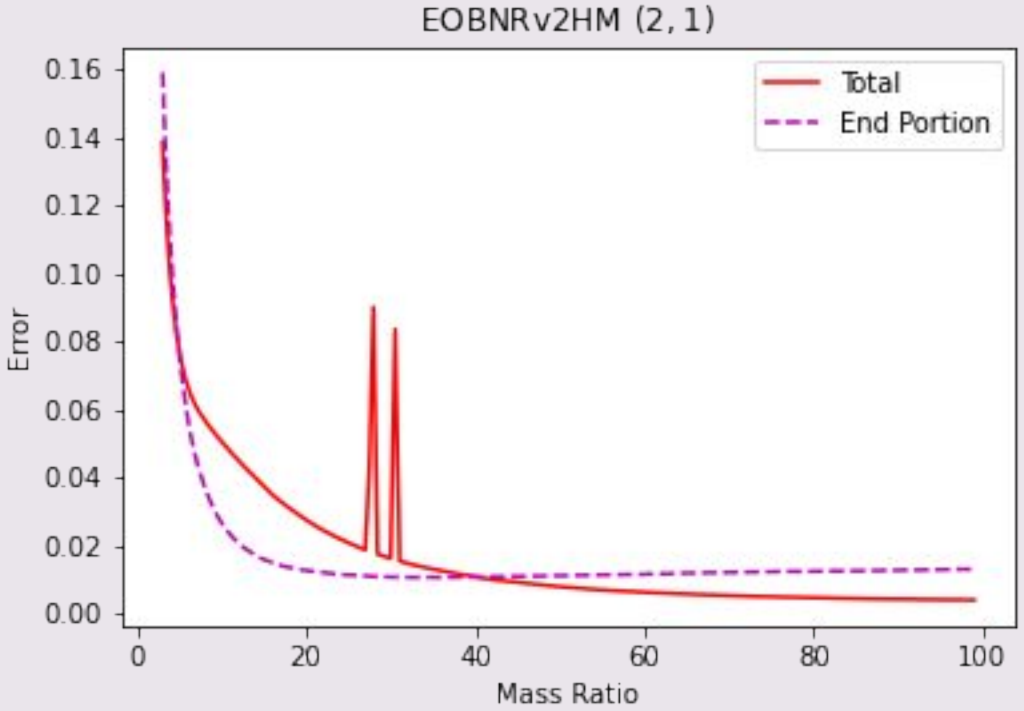


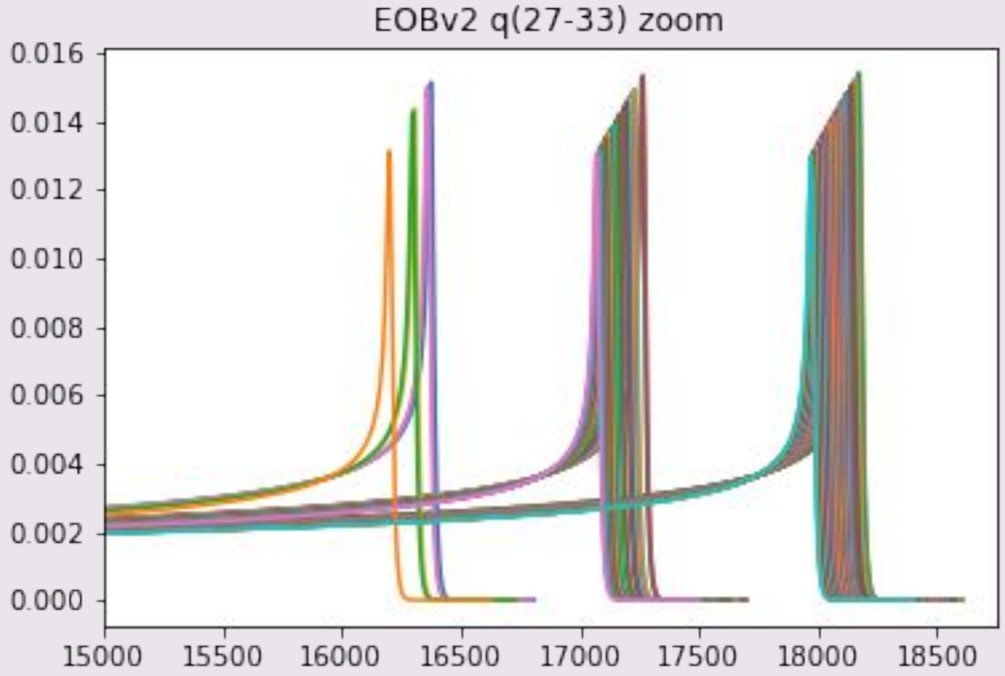
EOB (2,1) q=271
EOB
h
Time



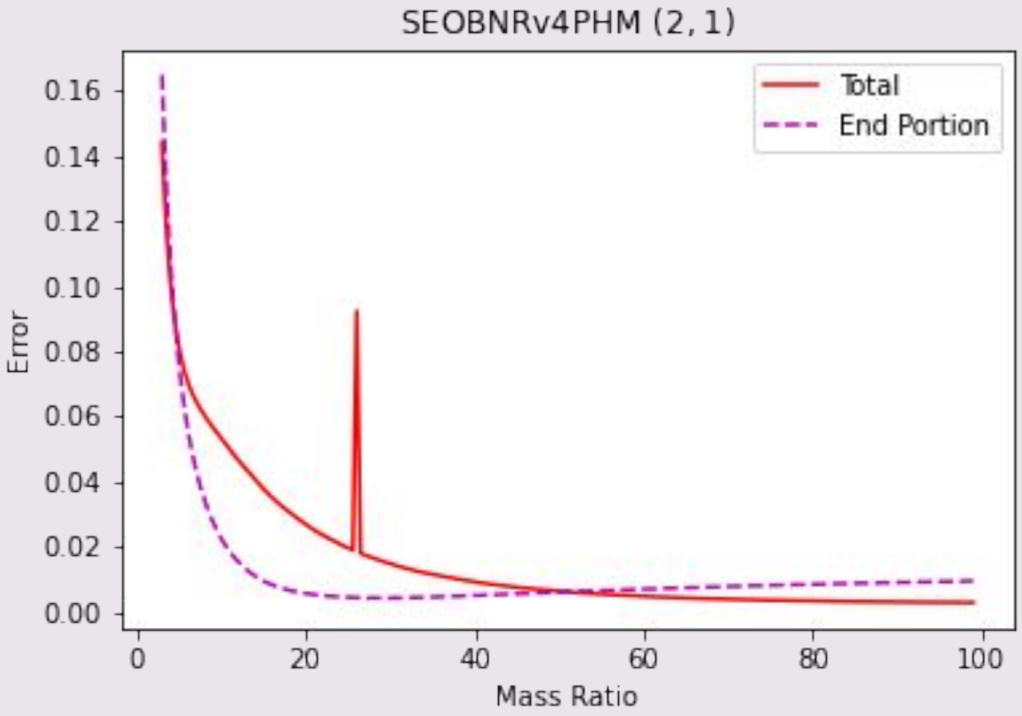
EMRI (2,1) q=271
EMRI

Results


EOBNRv2HM (2, 1)
Total
End Portion
Error
Mass Ratio
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
20
40
60
80
100


EOBv2 q(27-33) zoom
0.016
0.014
0.012
0.010
0.008
0.006
0.004
0.002
0.000
15000
15500
16000
16500
17000
17500
18000
18500

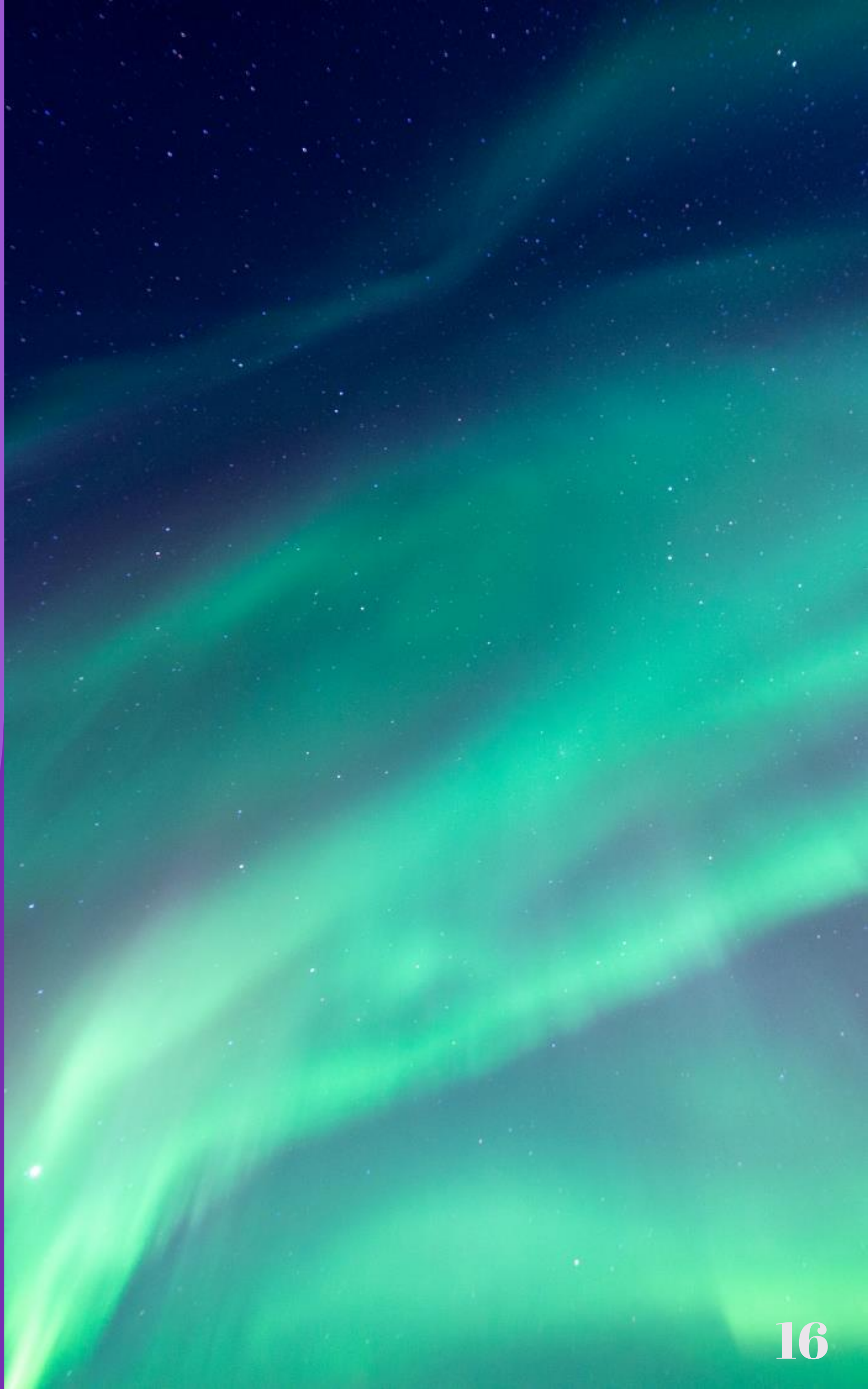
Results



- SEOBNR – Most recent EOB model
- Restricted to q < 100
- Similar results to EOB

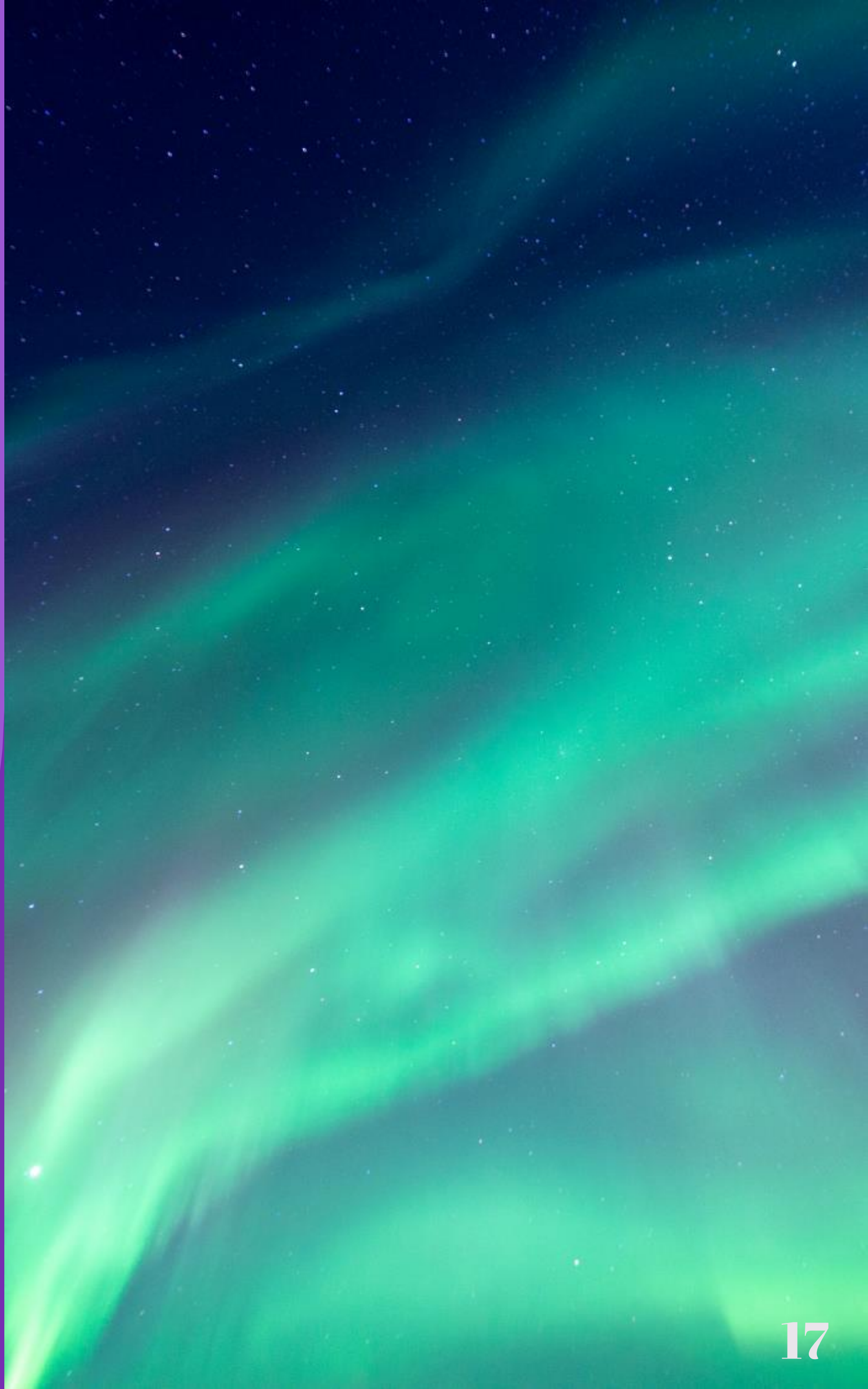
Results Conclusion

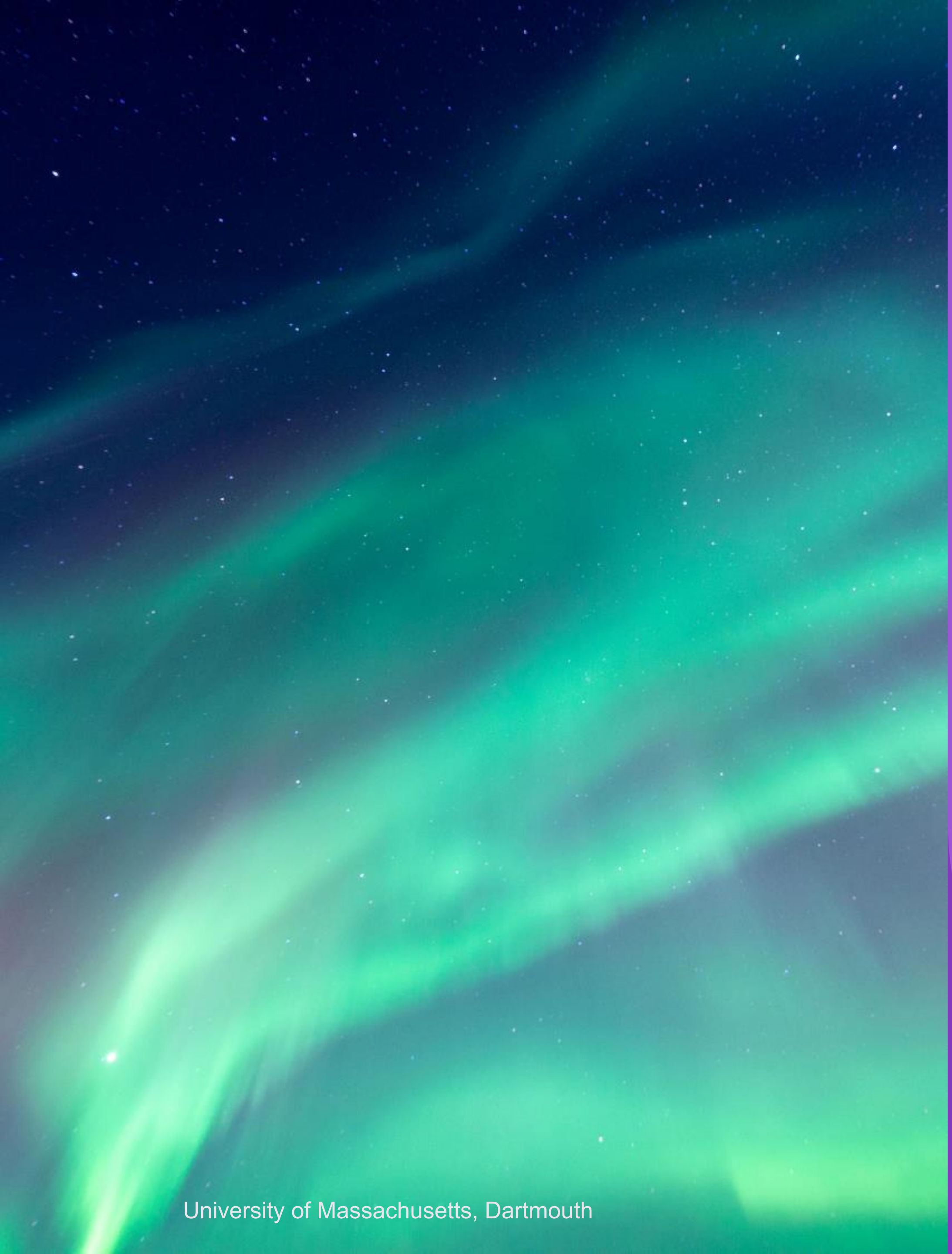
- EMRI and EOB agree/disagree at various mass ratios
- EMRI computation times faster
  - Even more so at higher mass ratios
- EOB has unexplained feature in (2,1) mode while $q > 100$



Looking to the Future

- Determine cause of shape of disagreement
    - Should be lower at low q values
- Minimize/Eliminate spikes in error
- Fix different lengths of EOB waveforms
- Cannot be sure which model is better
    - Einstein's equations unsolved in this mass ratio regime
    - Assessing model accuracy in this mass ratio regime is an open problem





Thank You

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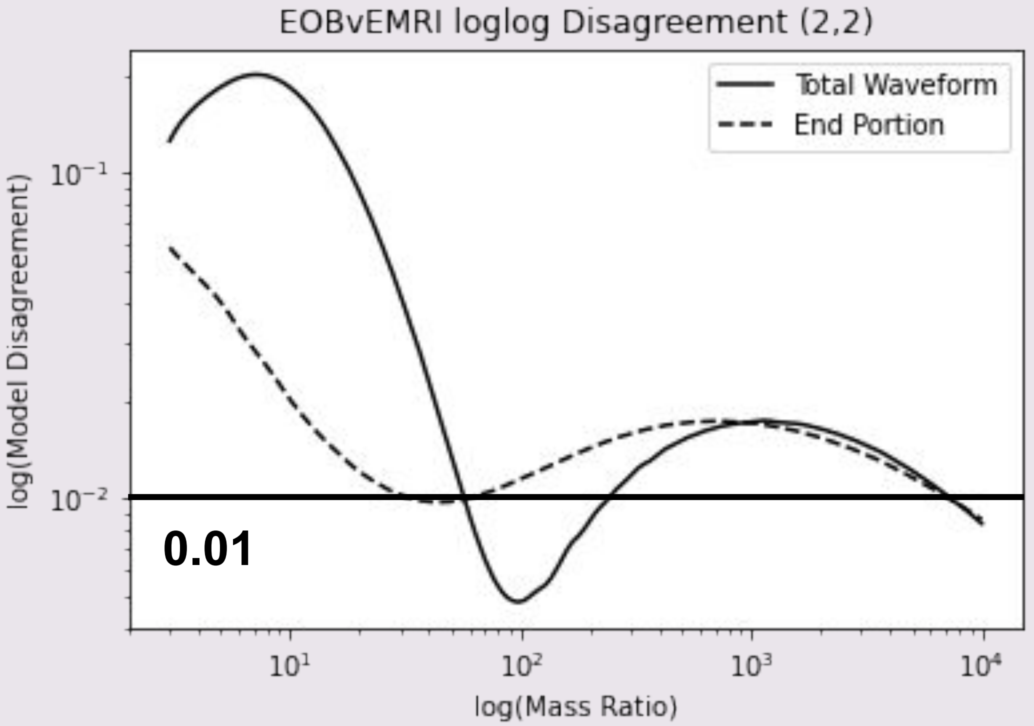
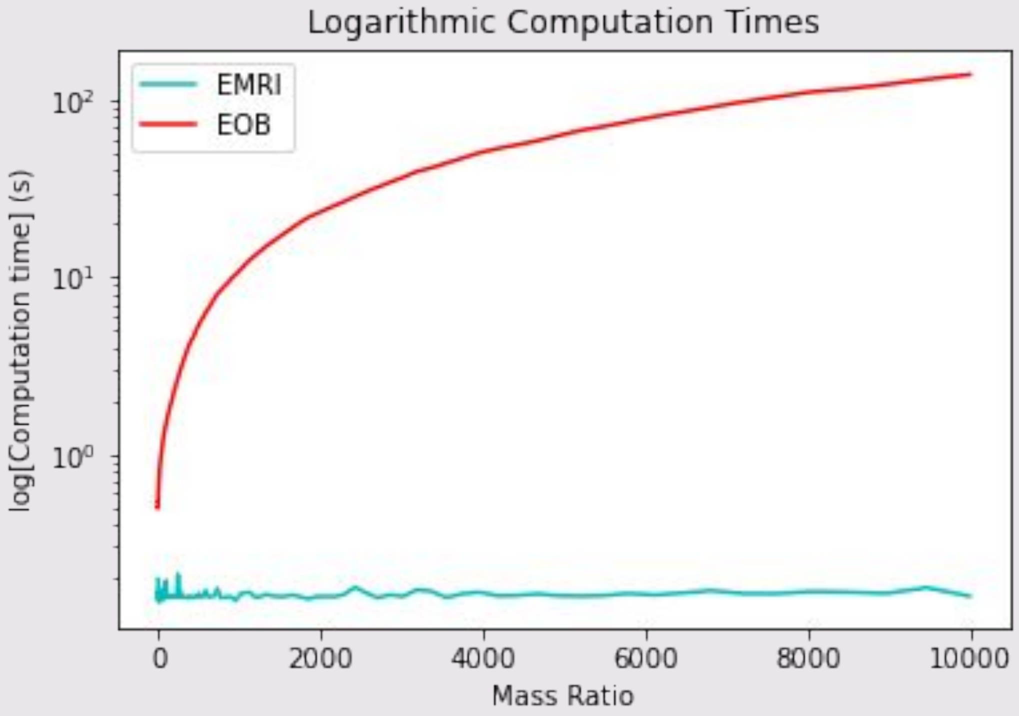
Mass Space Grant Consortium

Recent Work
(post 2021-URC)

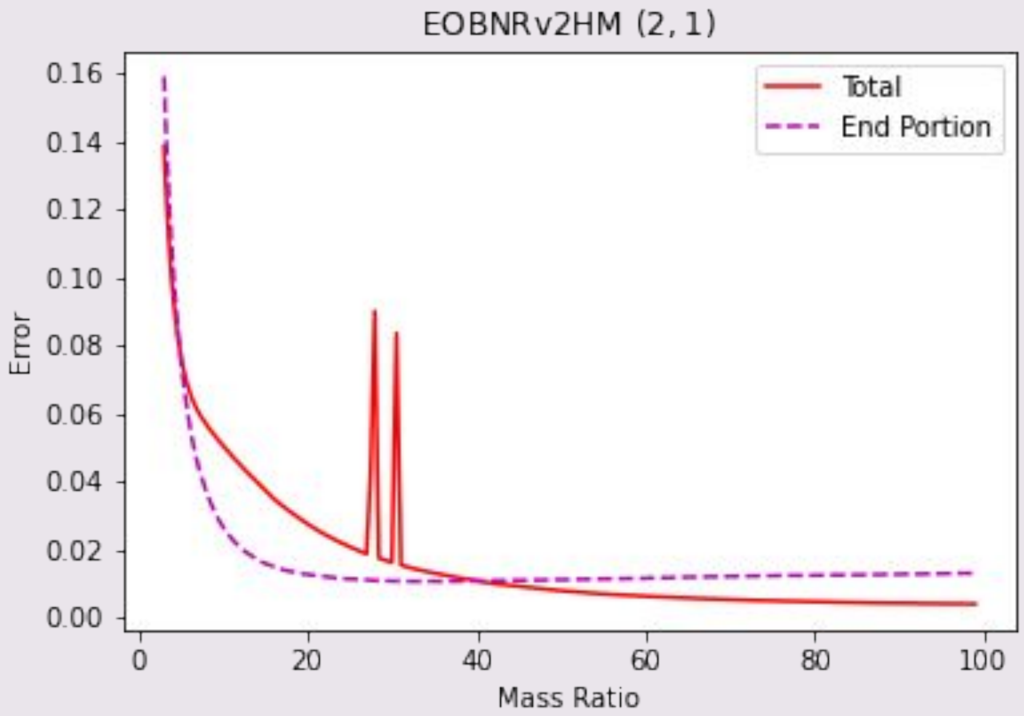
Results after spike bug fix



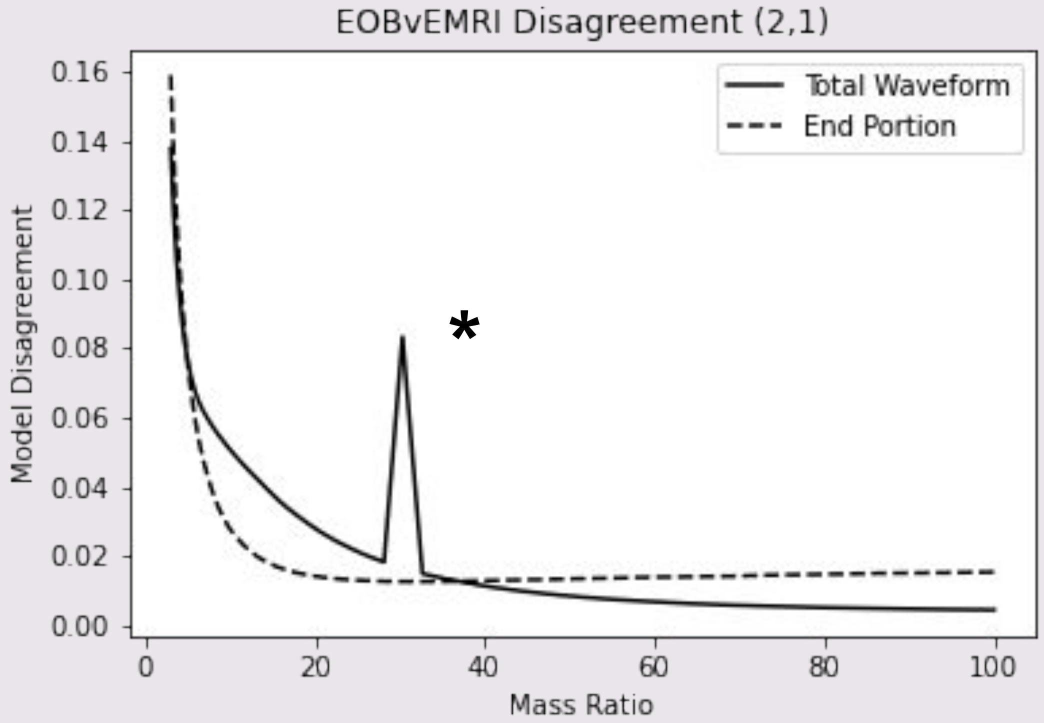


Results after spike bug fix

Before:

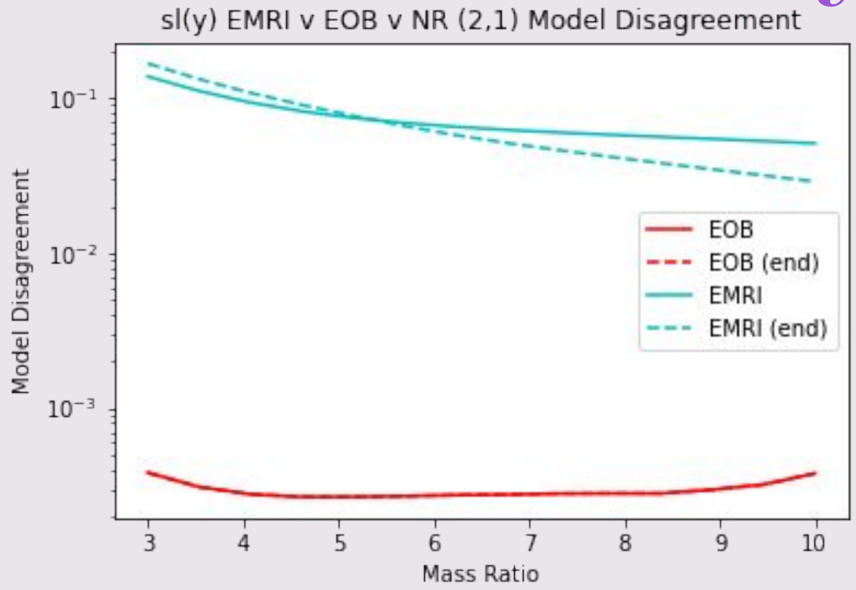


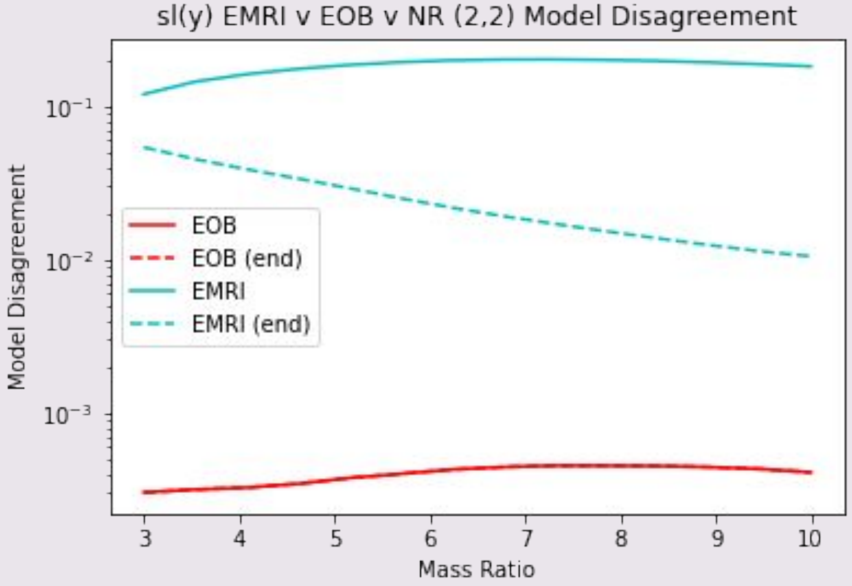
After:

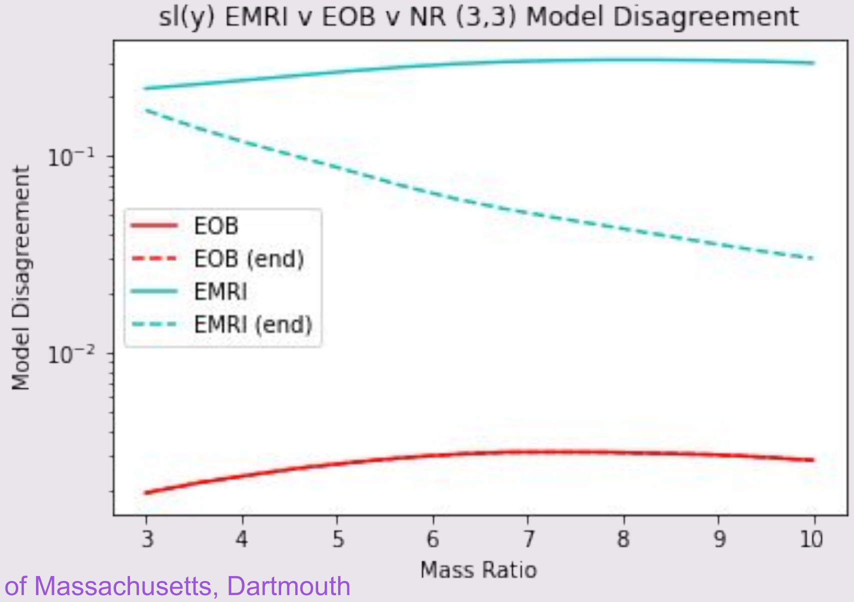


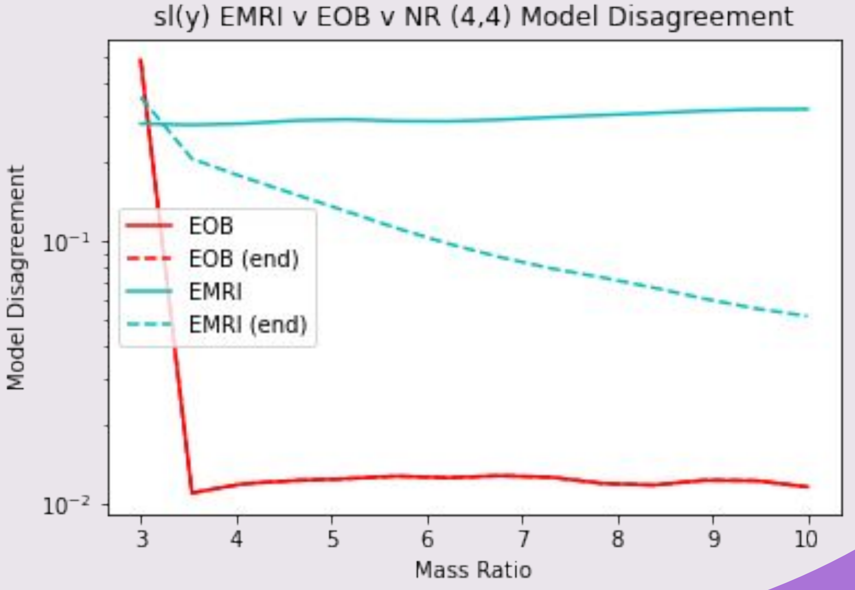
*It should be noted that the bug was related to variables using float32 instead of float64 data types. Many variables were changed to 64, this indicates there is still at least one variable in 32 format. All the variables will be fixed in the EMRI surrogate update within a few months.

Numerical Relativity Comparisons









Link to Project Notebooks and Figures

https://umassd-my.sharepoint.com/:f:/g/personal/kcullen_umassd_edu/En9E_1B2uOJNmQayfNQQrqAB1qjD2lNwI7CeFIOZxNICww?e=iqxJbc

This OneDrive file can be accessed by anyone within the University of Massachusetts, Dartmouth system (umassd.edu email)